Akhil Deshpande

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Software engineer with 2.5 years at Visa building payment-routing microservices and internal developer tooling in Java and Python, with hands-on experience in Spring Boot, Kafka, MSSQL, Kubernetes, and production on-call.

Experience

Software Engineer 08/2024 – Present
Visa Inc. *Austin, TX*

- Designed the REST endpoints, payloads, and error codes for a Java Spring Boot payment-routing and retry microservice (5,000 requests/day) with a senior engineer, doubling the retry success rate over the legacy framework
- Owned the MSSQL data layer and API for a cross-team migration of routing and retry rules to a new framework (schema design, migrations, replication), shipping 3 production releases on time and retiring the legacy system
- Created a React UI for the new routing API with lazy loading and pagination, giving teams more flexibility to manage routing and retry rules
- Engineered test frameworks for CyberSource APIs, automating SSH, Kafka messaging, and database validation
- Cut testing time by about 20% by eliminating redundant test runs using JaCoCo coverage data and Python analysis
- Developed an internal search tool (Python, Flask, Electron) across 4,000 CSV, config, and test-case files (1M+ LOC), saving about 2 hours per week per user for 40 engineers across 5+ teams
- Deployed Dockerized microservices to Kubernetes via Argo CD and Jenkins CI/CD; served in the on-call rotation, triaging production issues with Splunk, Grafana, and ClickHouse
- Mentored 2 new hires and 2 interns (both interns received return offers) and performed code reviews for the team

Software Engineering Intern 05/2023 – 09/2023
Visa Inc. *Austin, TX*

- Built a Flask web app in Python that lets engineers sign in, select a test environment, and send payloads via Kafka to a phone number or email
- Automated end-to-end verification with Paramiko (SSH), confirming each entry landed in the Oracle database

Software Engineering Intern 05/2022 – 08/2022
Dexcent Inc. *Edmonton, AB*

- Built the test suite from scratch for a Django REST server with pytest, raising coverage from 0% to 93%
- Automated API testing with Postman, running 30+ requests daily to validate endpoints and debug server issues

Undergraduate Teaching Assistant – CS 303E: Elements of Computers and Programming 09/2022 – 05/2024
University of Texas at Austin *Austin, TX*

- Led weekly labs for 30+ students and graded 60+ assignments per week, working with the professor and 15 TAs to address common difficulties

Projects

PlanSpacer | *Python, Flask, Google Maps API, JS* 07/2024 – 09/2024

- Developed a Flask web app that suggests equitable activity locations for groups separated by large distances, ranking candidates with a custom heuristic on each member's Google Maps distance and driving time

Text Yourself Python | *Python, GraphQL, REST, GitHub Actions* 01/2024 – 06/2026

- Built a Python tool that texts twice-daily task and weather updates using the LeetCode GraphQL API and REST APIs, scheduled as a GitHub Actions cron job

Fixify | *Python, Flask, CSS, HTML, JS, MySQL* 03/2023 – 05/2023

- Built a Flask app that scans users' Spotify playlists and identifies songs that have been removed and are no longer playable
- Cached per-user data in a MySQL login database and added an LRU cache to reduce computational load

Technical Skills

Languages & Frontend: Java, Python, JavaScript, SQL, HTML/CSS, React, Electron
Backend & Data: REST APIs, Spring Boot, Flask, Django REST Framework, Kafka, MSSQL, Oracle, MySQL, ClickHouse
DevOps & Monitoring: Docker, Kubernetes, Jenkins, Argo CD, GitHub Actions, Git, Splunk, Grafana
Testing & Data Tools: JaCoCo, Postman, pytest, Paramiko, NumPy, pandas

Education

University of Texas at Austin Austin, TX
Bachelor of Science in Physics *08/2021 – 05/2024*

- **Relevant Coursework:** Elements of Software Design (CS 313E), Discrete Mathematics

Product School Austin, TX
Product Management Certification *07/2024 – 11/2024*